

A DATA-DRIVEN EXPLORATION OF QUR'ANIC ROOTS

From Frequency to Structure



Measuring how the Qur'an's ~1,700 roots behave across all 6,236 ayahs — and reading those measurements honestly.

State a computed **FACT**, then a separately labeled **INTERPRETATION** — never blur them.

WEEK 1

Frequency

How often does a root appear — and how honestly can we read that?

The simplest, most objective layer. Worked roots: عدل vs ظلم.

Every value computed from Book6 (6,236 ayahs · ~1,700 roots).

The needle doesn't feel the earthquake



A seismograph draws a jagged line. Hand the same paper to anyone — they read the same peaks.

But the needle never felt the fear, never saw a wall fall.

It recorded one thing only: that the ground moved — and how much.

That is all a count is.

This term we are the needle. We mark where a word shakes the page —
not what the trembling meant to those who lived it.

A reading can be wrong, so it can be re-read. A feeling can only be argued.

We hand you the reading. The meaning stays yours.

The frame & today's question

If I asked which idea matters most in the Qur'an, you'd answer from memory — and we'd have no way to check each other.

**So we take a smaller, humbler question we CAN settle:
not what matters most, but how often a root appears.**

The course's one rule, kept all term:

state a computed FACT, then a separately labeled INTERPRETATION — never blur them.

We measure not to replace reading, but to make claims about the text AUDITABLE:

a count can be wrong, but it can be checked.

Today we claim presence only — not importance, not meaning, not relationships.

What frequency analysis is

Counting how often a meaning-bearing unit occurs in a text —

the word-frequency, or “bag-of-words,” tradition in corpus linguistics.

Temporarily set aside order and grammar; ask only: what is present, and how often?

Like emptying a book into a bag and counting the words —

you lose the sentences, but you can finally tally what's there.

The working assumption is modest:

how often a text names something is a SIGNAL of emphasis — a trace, not proof.

We start here because it's the most objective, reproducible layer;

everything later — distribution, partners, co-occurrence — builds on these counts.

From text to countable units

The same idea wears many clothes; we want to count the idea, not the spelling.

The pipeline:

- Tokenize each ayah into words.
- Reduce each word to its 3-consonant ROOT: كَتَبَ → كِتَابُ / يَكْتُبُ / كَتَبَ.
- Normalize letters & diacritics so the same root matches everywhere.
- Drop function words (في، من، الذي) — they carry grammar, not theme.

Count once per ayah (document frequency) — so one long verse can't dominate by repeating a root.

The reduction is lossy ON PURPOSE — we trade nuance for comparability.

After it, the corpus holds 51,024 root-tokens.

Normalization to a rate – 1 Not stark

A bigger text yields bigger counts regardless of emphasis —

so raw counts can't be compared. Convert to a RATE.

- per 1,000 AYAHS = $\text{ayah-freq} \div 6,236 \times 1,000$ (share of verses).
- per 1,000 ROOTS = $\text{term-freq} \div 51,024 \times 1,000$ (size-true).

Why two? The denominator can flip the ranking — that's not cosmetic:

هدي < ظلم per ayah (46.5 vs 43.0), but ظلم < هدي per root (6.19 vs 6.17).

Per-1,000-roots is like deaths-per-100,000 instead of raw deaths — the only fair way to compare populations of different sizes.

Always report both rates: presence is a RATE, not a count.

Normalization to a rate – 2 Per Ayah Stark

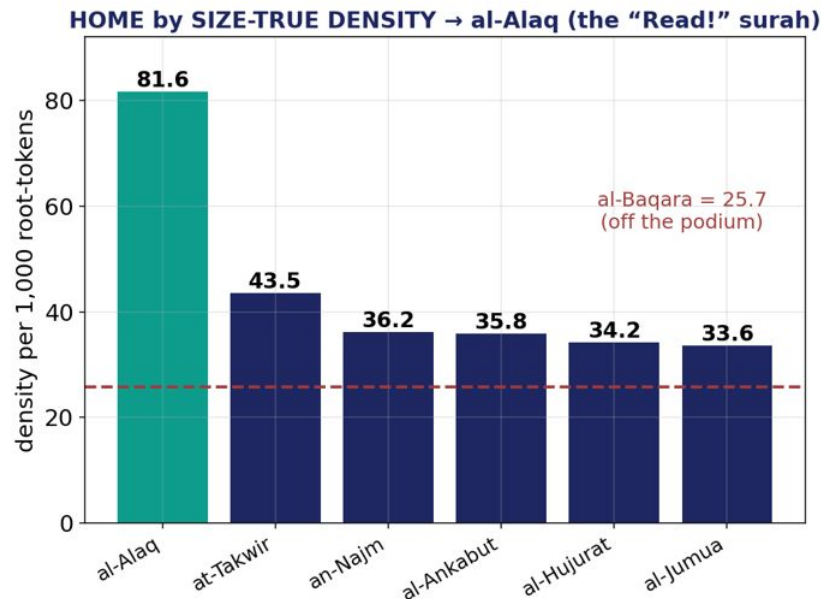
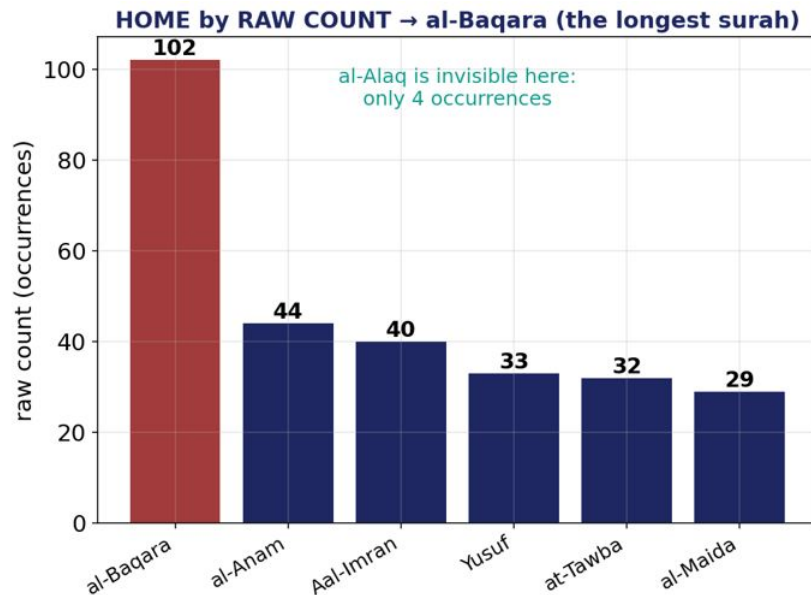
Within-verse repetition:

- **وحد Oneness:** 23.7 per 1,000 ayahs (the most verses) → but only **3.00** per 1,000 roots → **last**
- **قتل kill:** 19.6 per 1,000 ayahs → **3.33** per 1,000 roots → **first** (war verses repeat it: قَاتِلُوا ... قَاتِلُوا)
- **بعض some:** just 14.0 per 1,000 ayahs (fewest verses) → leaps to **3.10** per 1,000 roots . repeat: 1.82 — بعضهم ... بعضاً

So oneness is named in the *most verses* yet drops to *last* once you count every occurrence — a real plunge, not a hairline.

Normalize before you compare; a count is where you start, a rate is what you trust.

Normalization to a rate – علم 3



LEFT - home by raw count: بقره towers at 102 occurrences, simply because it's the longest surah. علق is invisible (4 occurrences).

RIGHT - home by size-true density (per 1,000 root-tokens): the ranking inverts. علق leads at 81.6, and بقره — sits at 25.7, off the podium (red dashed line).

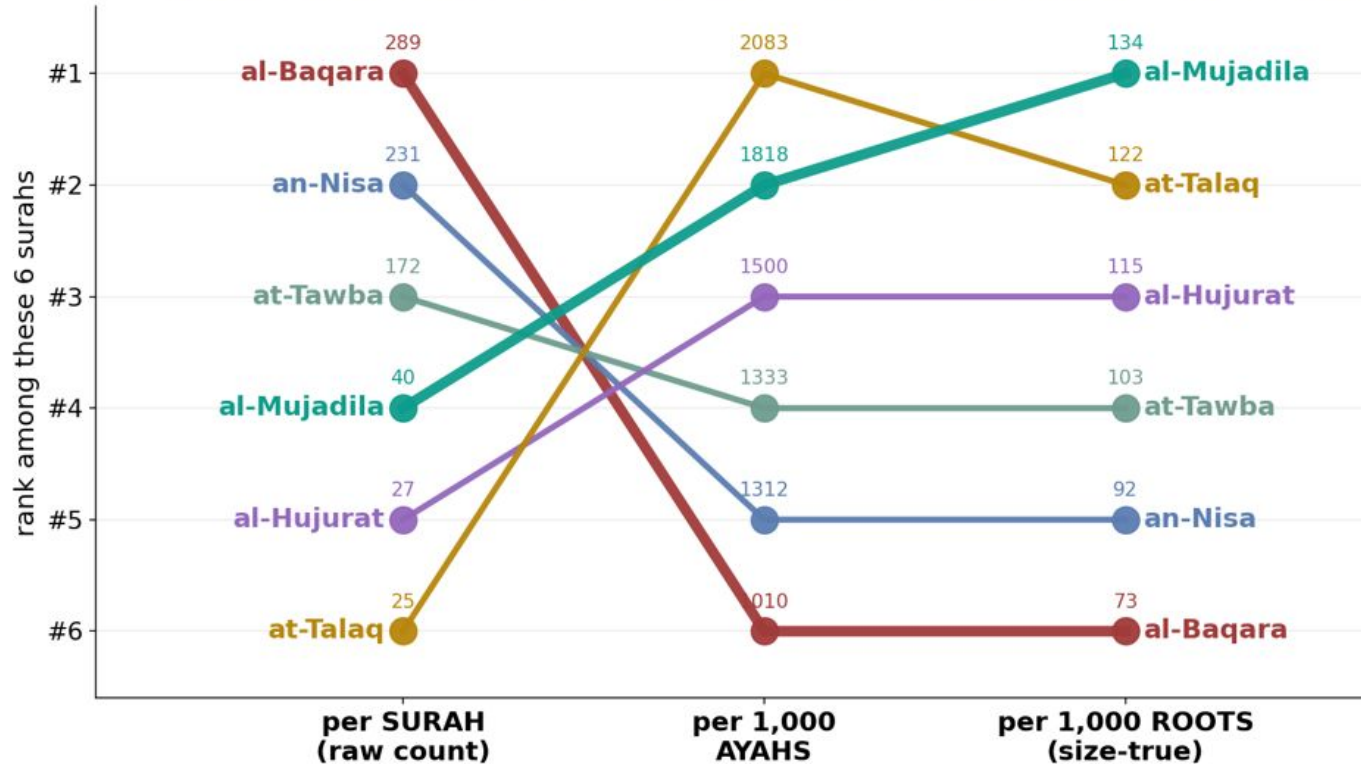
Normalization to a rate – 4 علم

THE TAKEAWAY lands hard because it's also meaningful:

- The size-true measure crowns علق, the first revelation
- "Read! ... who taught by the pen" — as the true home of "knowledge,"
- Exactly where a human reader would put it
- The raw count just rewards the biggest surah
- "presence is a rate, not a count," made stark.

Normalization to a rate – 5 عله

“Allah” (Allāh): the home surah reshuffles by normalization
al-Baqara plunges #1 → last; at-Talaq & al-Mujadila rise to the top



Normalization to a rate – 5a **ءله (freq=2848)**

Surah	root ءله (raw)	÷ ayahs → per 1,000	÷ roots → per 1,000
al-Baqara	289	289/286 × 1000 = 1010	289/3965 × 1000 = 73
an-Nisa	231	231/176 = 1312	231/2496 = 92
at-Tawba	172	172/129 = 1333	172/1666 = 103
al-Mujadila	40	40/22 = 1818	40/299 = 134
al-Hujurat	27	27/18 = 1500	27/234 = 115
at-Talaq	25	25/12 = 2083	25/204 = 122

ریشه	توکن ریشه	ریشه
2,697	الله	ءله
5	اللهم	ءله
95	إله	ءله
16	إلهها	ءله
33	آلهة	ءله
2	إلهين	ءله
2,848	جمع	جمع

Normalization to a rate – 6 عله

Ranking the same 6 surahs in each column

- بقره (red) plunges #1 → #6 → #6: first by raw count (289), dead last once you correct for size. The longest surah's dominance is pure container effect.
- طلاق rises #6 → #1 → #2: invisible by raw count (25), but the densest per ayah.
- مجادله (teal) climbs #4 → #2 → #1: the size-true winner — the surah where "Allah" sits in every verse.
- **Three measures, three different #1s, and the crossing lines make the reshuffle unmistakable**

The themed-root frequency ladder

Rank a themed set of roots by rate — the shape of the vocabulary appears.

Three descriptive bands:

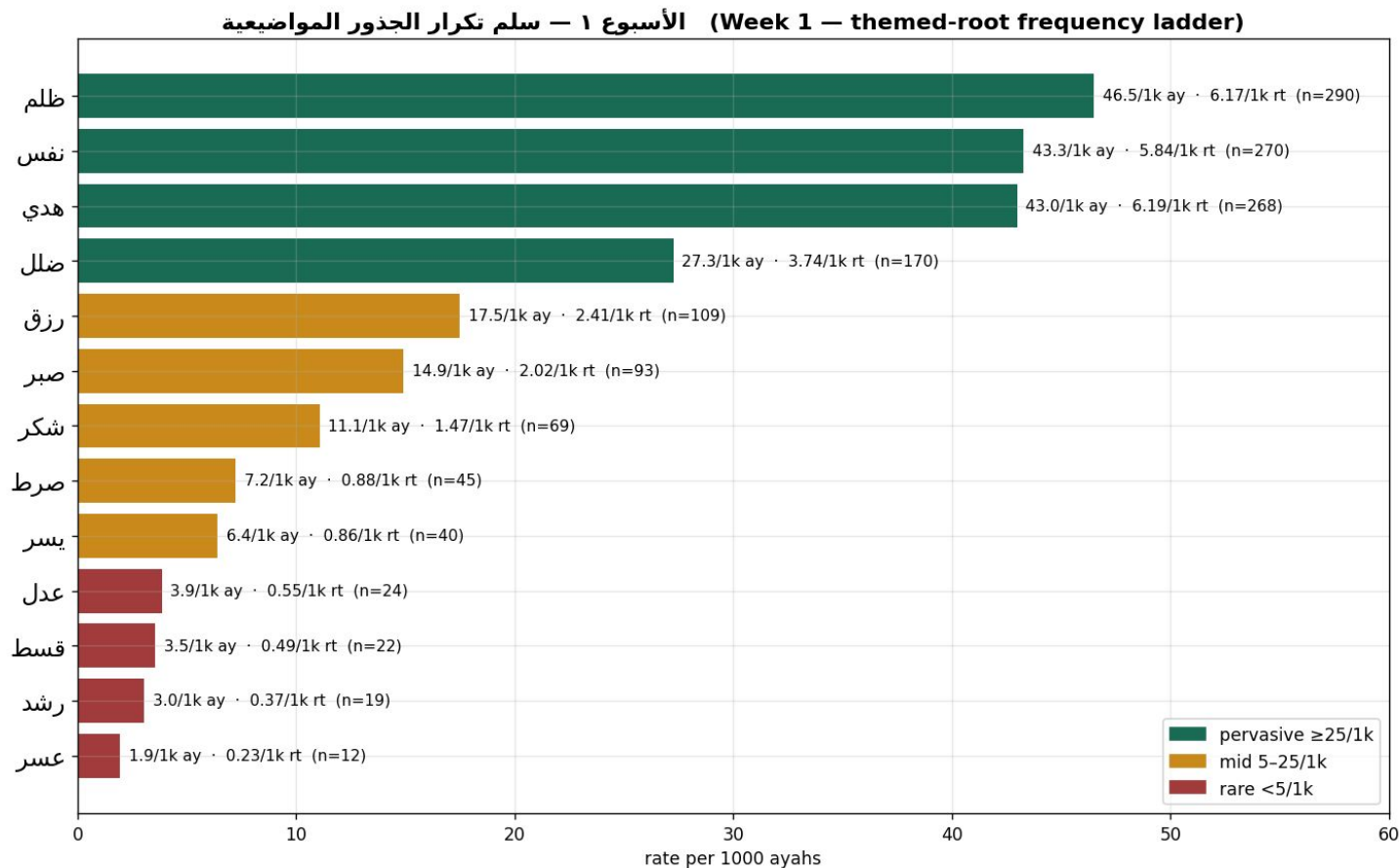
- pervasive ($\geq 25/1k$): ظلم، نفس، هدي، ضلل
- mid (5–25): رزق، صبر، شكر
- rare (< 5): عدل، قسط، عسر

The bands orient attention — they are not a law of the text.

Live in the app: 465 → كفر ayahs (the most-named); 12 → عسر (the rarest).

Always pair a rate with its raw count and its size-true rate.

The themed-root frequency ladder



Predict, then check — ease vs hardship

Before we look — which is named more: ease (يسر) or hardship (عسر)?

Most rooms guess hardship.

Check it:

40 — يسر ayahs.

12 — عسر ayahs.

About 3.3× more “ease.”

The habit you just watched: pair the rate with the raw count BEFORE you speak.

And one number is about to unsettle a far more common assumption...

Live in the app — the Per-Root Profile

Type a root, press Analyze, open the Per-Root Profile. For ظلم you get:

- 290 ayahs matched, 315 total, 17 surface forms.
- 98.59th percentile — more frequent than ~98.6% of all roots.
- tier: “ubiquitous” (top 1%).
- first occurrence 2:17 (بقره); last 76:31 (الانسان).

Compute the two rates right there:

46.5 per 1,000 ayahs (share of verses) · 6.17 per 1,000 roots (size-true).

Reading habit: always pair the percentile with the raw count and BOTH rates before you say a single word about the root.

ریشه	ظلم	ظلم	ظلم	ظلم	ظلم	ظلم	ظلم	ظلم	ظلم	ظلم	ظلم	ظلم	ظلم	ظلم	ظلم	ظلم	ظلم	جمع
توکن ریشه	أظلم	تظلم	ظالم	ظالمة	ظالمون	ظالمی	ظالمین	ظلام	ظلم	ظلما	ظلمات	ظلوم	ظلوما	مظلما	مظلمون	مظلوما	يظلم	
تعداد	17	9	5	4	33	2	91	5	77	8	23	1	1	1	1	1	36	315

The headline finding & the “unlearn”

Predict with me: which does the Qur'an name more —
عدل (justice) or ظلم (injustice)?

Most rooms say justice. The data says otherwise:

290 — ظلم ayahs (98.6th percentile, “ubiquitous”).

24 — عدل ayahs.

About 12× more.

A genuine unlearning: the corpus names the VIOLATION far more than the IDEAL.

But stay disciplined — this is presence, not endorsement: ظلم is named to condemn it.

Frequency as a vector [preview]

Here is the idea that makes today bigger than it looks.

List a root's counts across contexts — that ordered list of numbers is a VECTOR.

Do it for every root, and each root becomes a POINT in a space;
roots with similar profiles sit close together.

That is the VECTOR SPACE MODEL — the foundation of how machines “read” text.

Everything later — partners, co-occurrence, networks — is geometry on these vectors.

Nothing new is computed here: the very counts we tabulated are the coordinates.

Labeled preview — embeddings & skip-gram are further-study, never asserted against Book6.

What frequency gives — and what it loses

GIVES:

- objective & reproducible — anyone re-running gets the same numbers.
- scalable to all 6,236 ayahs.
- comparable, via rates.
- scoping — shows WHERE to look deeper.

LOSES — context, above all:

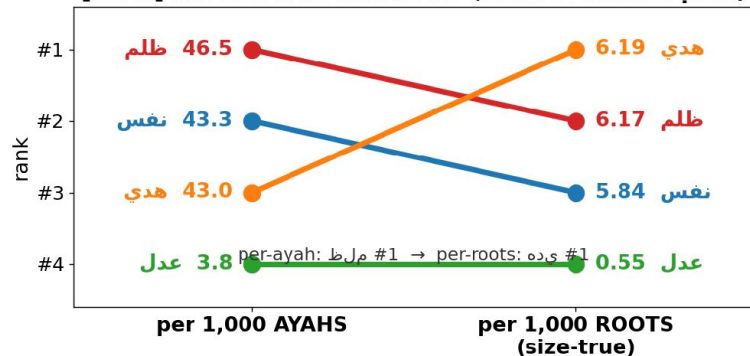
- stance/polarity (كفر counted whether affirmed or refuted).
- polysemy (ظلم = injustice AND ظلمات = darkness, as one).
- speaker, syntax, and all relationships between roots.

Frequency **SCOPES** inquiry; it does not **SETTLE** it.

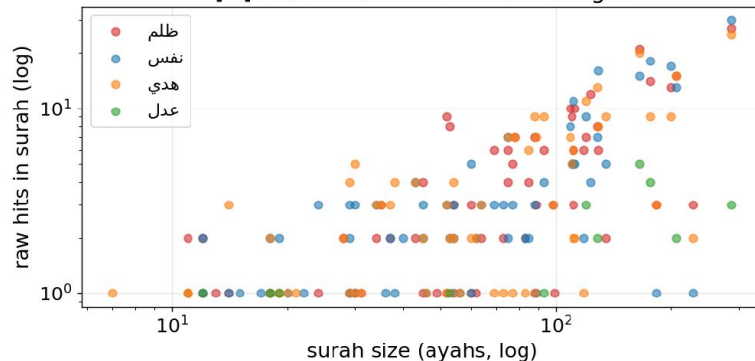
What frequency gives — and what it loses

الأسبوع ١ — جوانب التكرار: ما يكشفه وأين يضلل (Week 1 — frequency: what it reveals & where it misleads)

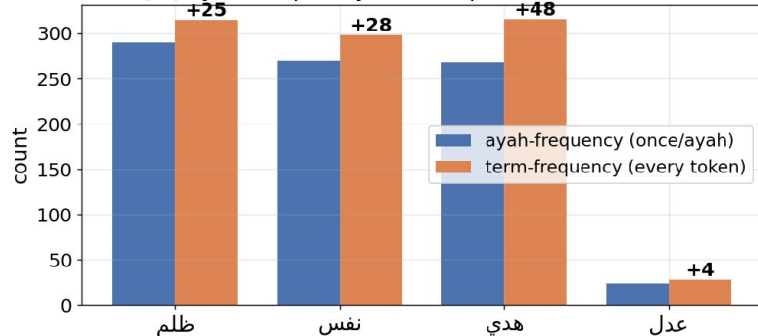
[norm] the size-true RANK FLIP (hudā overtakes zulm)



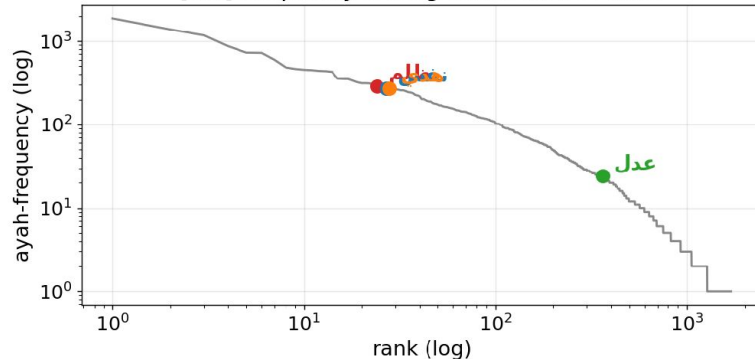
[–] raw count rides on surah length



[–] ayah-frequency hides repeats inside a verse



[ctx] frequency is long-tailed (1700 roots)



What frequency gives — and what it loses

Root	Gloss	Ayahs (n)	/1k ayahs	/1k roots	Band
كفر	disbelief	465	74.6	10.29	pervasive
ظلم	injustice	290	46.5	6.17	pervasive
هدي	guidance	268	43.0	6.19	pervasive
ضلل	going astray	170	27.3	3.74	pervasive
صبر	patience	93	14.9	2.02	mid
شكر	gratitude	69	11.1	1.47	mid
يسر	ease	40	6.4	0.86	mid
عدل	justice	24	3.8	0.55	rare
عسر	hardship	12	1.9	0.24	rare

Fact vs. interpretation — the discipline

Separate the computed fact from any reading of it.

One sentence of fact (with its normalization), one labeled sentence of interpretation.

FACT

“ظلم is named in 290 ayahs (46.5/1k ayahs, 6.17/1k roots) — about 12× عدل.”

INTERPRETATION

“I read this as the text foregrounding the diagnosis of wrong over the abstract ideal.”

Peer test: can a classmate tell which sentence is fact and which is interpretation?

If not — rewrite until they separate cleanly.

Worked example — al-Fatiha (by hand)

Compute frequency by hand on 7 ayahs:

tokenize → roots → count by ayah, then by token.

The teaching point — repetition:

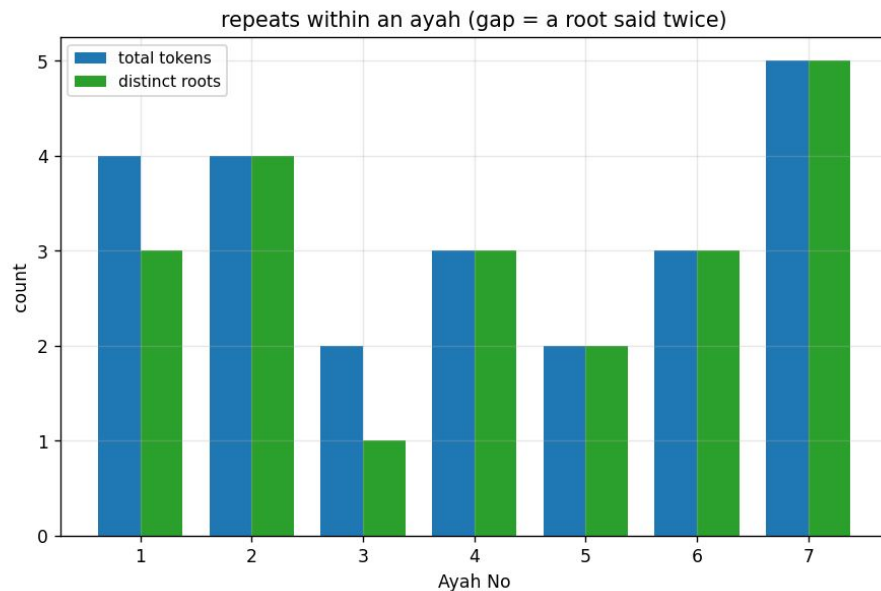
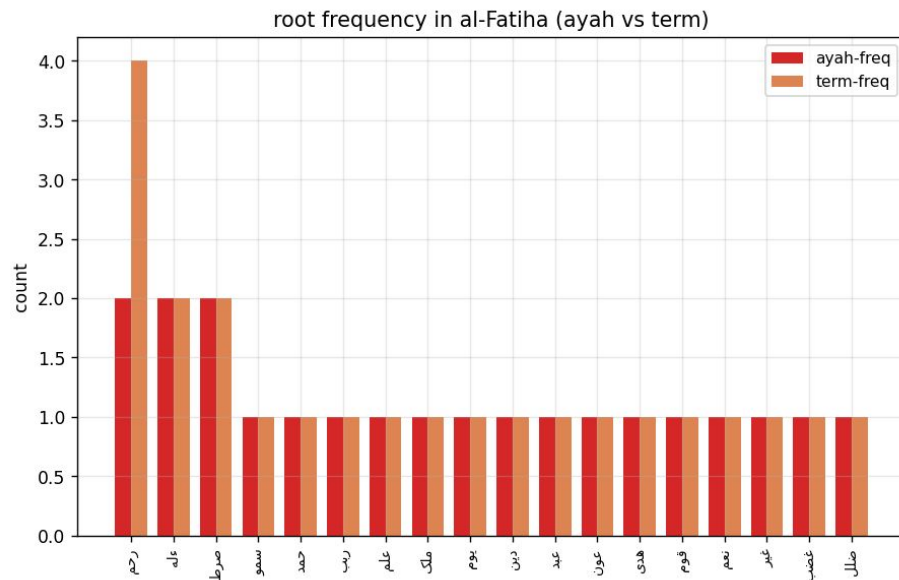
رحم has ayah-frequency 2 but term-frequency 4 —
the only root uttered four times.

Ayah-counting hides repetition inside a verse;
term-counting catches it.

Then: one fact (“رحم, ayah-freq 2, term-freq 4”)
+ one labeled interpretation.

Worked example — فاتحه (by hand)

سورة الفاتحة — مثال محسوب يدويا للتكرار (al-Fatiha — worked example: frequency)



This week's exercise

Part 1 — by hand

- Count the roots of your assigned short surah, by ayah and by token.
- Flag any root whose term-frequency exceeds its ayah-frequency.

Part 2 — in the app

- Profile your assigned root; record BOTH rates (per 1K ayahs AND per 1K roots).
- Note percentile and tier; take one screenshot.

Submit one screenshot + two readings (one fact + one interpretation each),
the night before class — it gates the debrief.

Three ways to never forget: presence is a RATE, not a count

Always ask: per how much?

Salt: pot vs. cup

- One tablespoon of salt makes a teacup undrinkable — but is barely tasted in a giant stockpot.
- al-Baqara names “Allah” 289× yet it’s a stockpot: diluted to 73 per 1,000 roots. al-Mujadila is the teacup — saturated at 134.

“It’s not how much salt — it’s how salty.”

Stadium vs. tiny club

- 500 of 50,000 sing along (1%); in a 50-person club, 40 sing (80%). The stadium has more voices, the club more devotion.
- The longest surah racks up the most mentions of everything — like the biggest crowd holds the most of every kind of person.

“The biggest crowd isn’t the most devoted — measure the %, not the headcount.”

Textbook vs. pamphlet

- A word highlighted 50× across a 500-page textbook vs. 4× in a 4-page pamphlet.
- The pamphlet is the one truly about that word — why “knowledge” lives in tiny al-‘Alaq, the “Read!” surah, not giant al-Baqara.

“Density, not tally — the small page can be the loudest.”

Presence is a rate, not a count — and it is context-blind

Frequency is the most objective layer — but the most context-blind.

It can defy intuition (ظلم 12 × عدل) — yet it never tells you why, who, or how.

Next week — distribution & concentration:

Frequency says HOW MUCH; next we ask WHERE a root sits and how it spreads.